

Molecular Epidemiology of Rifampicin-Resistant *Mycobacterium tuberculosis* by GeneXpert MTB/RIF Assay in Renal Transplant Recipients: A Single-Center Experience

Minaxi H. Patel,¹ Kamla Kanodia,² Lovelesh Nigam,² Rashmi Patel,² Kamlesh Suthar,²

Himanshu Patel,³ Vivek Kute,³ Divyesh Engineer,³ Sanskriti Chauhan,³

Hari Shankar Meshram³

Abstract

Objectives: There are scarce data on the incidence and resistance pattern of rifampicin-resistant *Mycobacterium tuberculosis* among kidney transplant recipients.

Materials and Methods: This is a retrospective, single-center study of kidney transplant recipients suspected of *M. tuberculosis* infection. The GeneXpert assay we used detected mutations in the *rpoB* gene that confer rifampicin resistance using 5 overlapping probes (A, B, C, D, and E). The probes can detect mutations in the codons 507 to 511 (probe A), 511 to 518 (probe B), 518 to 523 (probe C), 523 to 529 (probe D), and 529 to 533 (probe E). We also detailed the treatment protocol and outcomes of kidney transplant recipients infected with rifampicin-resistant *M. tuberculosis*.

Results: In total, 2700 samples were processed during the period from October 2018 to February 2022 with successful results in 2640 samples (97.04%). One hundred and ninety (7.19%) samples were positive for *M. tuberculosis*, and rifampicin resistance was detected in 12 (0.45%) cases (11 pulmonary, 1 genitourinary). The most common *rpoB* mutation was located in the region of probe E (75.0%), followed by probe A (16.6%) and in 1 combination probe DE (8.33%). The *rpoB* mutations were not observed in probe B and probe C. Six patients received bedaquiline-based treatment for a short course of 11 months, whereas the other 6 patients required a long course of 18 to 20 months.

Three patients died, 2 were lost to follow-up, and 7 were cured. During treatment, 4 patients experienced acute rejection, and 1 graft loss was reported.

Conclusions: We report for the first time the incidence and pattern of rifampicin resistance among kidney transplant recipients with tuberculosis infection. Further investigations are required for exploring the molecular and clinical phenotypes.

Key words: Biosafety, Isoniazid resistance, Nucleic acid amplification, Tuberculosis infection

Introduction

Kidney transplant recipients (KTR) have an increased risk for tuberculosis (TB) infection compared with the general population, and the prevalence of TB in KTR patients varies widely geographically.¹ The diagnosis of TB in KTR patients is difficult,² and treatment leads to high risks of drug interactions and toxicity,³ which increase the risk of graft dysfunction/graft loss.⁴ India has a high incidence of multidrug-resistant tuberculosis (MDR-TB), which is emerging as an obstacle to effective control of TB. Rifampicin is considered an important drug in anti-TB treatment because of its early bactericidal effect on metabolically active bacteria and excellent late sterilization action on semi-dormant organisms.⁵

Multidrug resistant tuberculosis is defined as TB resistant to at least 2 important first-line drugs, isoniazid and rifampicin. More than 90% of rifampicin-resistant (RR) TB cases are also associated with isoniazid resistance. Rifampicin resistance is of particular epidemiologic importance; therefore, it represents a valuable surrogate marker for MDR-TB. Infection by RR-TB requires long therapy with less effective and more toxic second-line drugs.⁶ Proper treatment of RR-TB relies on prompt diagnosis.⁷

From the ¹Microbiology Section, the ²Department of Pathology, Laboratory Medicine, Transfusion Services and Immunohematology, and the ³Department of Nephrology and Kidney Transplant, Institute of Kidney Diseases and Research Centre and Institutes of Transplantation Science, Civil Hospital, Ahmedabad, Gujarat, India

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Corresponding author: Himanshu H. Patel, Department of Nephrology and Kidney Transplant, Institute of Kidney Diseases and Research Centre and Institutes of Transplantation Science (IKDRC-ITS), Civil Hospital campus, Ahmedabad, Gujarat, India, 380016

Phone: +91 9727882222 E-mail: drhpatel@gmail.com

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Diagnosis of RR-TB has been traditionally difficult because it requires sophisticated biosafety and laboratory infrastructure. The World Health Organization has endorsed the GeneXpert assay for use as a point-of care molecular assay; simultaneous diagnosis of TB and RR is possible within 2 hours.⁸

Large-scale studies on RR-TB epidemiology among KTR patients are few; hence, we described our experience with the GeneXpert assay for determination of the molecular epidemiology of RR-TB at our transplant center.

Materials and Methods

Design, settings, and study population

This retrospective study was conducted at the Microbiology section in the Department of Clinical Pathology, Laboratory Medicine, Transfusion Services and Immunohematology. Data were collected from the hospital database of the Institute of Kidney Diseases and Research Centre and Institutes of Transplantation Science, Civil Hospital campus, Ahmedabad, Gujarat, India, for the period from October 2018 to February 2022.

The study participants included all KTR patients at our center on follow-up, irrespective of age and sex, suspected to have TB. The study was approved by the institutional ethical review board committee. Because deidentified data were used retrospectively, which preserves anonymity of the participants, no written informed consent was taken. All the transplants performed in the center are in accordance with the rules of regulations of the Declaration of Helsinki, the Declaration of Istanbul, good clinical practice, and the Indian Organ Act. Most of the transplants were from living related donors (82%), with some from deceased donors (18%). The relationships of living related donors included 51% spouses, 38% parents, 10% siblings, and 1% children.

GeneXpert assay description

All samples were processed by GeneXpert assay, which is a cartridge-based nucleic acid amplification test. The technique uses an automated hemi-nested real-time polymerase chain reaction and molecular beacon technology and is able to simultaneously detect the presence of MTB and as RR. The assay was performed using version G4 according to manufacturer's instructions (Cepheid). The GeneXpert assay uses 5 overlapping molecular probes

labeled A through E, which are complementary to the codons 507 to 533 located in the 81-bp rifampicin resistance determinant region in the *rpoB* gene as follows: probe A (507 to 512), probe B (512 to 518), probe C (518 to 523), probe D (523 to 529), and probe E (529 to 533). The World Health Organization has endorsed this assay for diagnosis of RR-TB in high-burden countries.⁹ Unsuccessful results were classified in 3 groups according to manufacturer's instructions: (1) invalid (failure of sample processing control due to the sample not properly processed or the polymerase chain reaction was inhibited), (2) error (failure of the probe check control due to the reaction tube filled improperly, or a reagent probe integrity problem, or the maximum pressure limits were exceeded, or there was a GeneXpert module failure), and (3) no result (the test was stopped due to power outage).^{10,11} We also collected the baseline details of the KTR patients, baseline immunosuppression, the treatment regimen selected, the change in treatment protocol, complications during treatment, and outcomes in terms of acute rejection, graft loss, and mortality.

Statistical analyses

No formal sample size calculation was performed for the study. No inferential statistical handling was applied to the data. All categorical data are presented as numbers (and percentages), and continuous data are shown as median values (with IQR). The statistical tool we used was STATA (version 16).

Results

During the study period, 2700 samples (both outpatients and inpatients) were processed. Demographic data of the study are summarized in Table 1. Overall, the assay provided a valid result in 2640 (97.04%) samples and an unsuccessful result in 80 (2.96%) samples. Of 80 unsuccessful results, 55 were classified as error, 20 as invalid, and 5 as no result due to power outages. We were able to retest 30 samples, and valid results with no MTB were obtained in 20 of 30 samples. Fifty samples could not be retested due to either insufficient quantity or unavailability of a second sample. Of 2640 results, MTB was detected in 190 (7.19%) samples.

Among the samples that were positive for MTB, 125 (65.8%) were pulmonary samples (65 sputum, 60 bronchoalveolar lavage fluid) and 65 (34.2%) were

extrapulmonary samples. Rifampicin resistance was detected in 12 (6.3%) samples: 11 pulmonary samples (9 bronchoalveolar lavage fluid, 2 sputum) and 1 extrapulmonary sample (urine) (Figure 1). An indeterminate result for rifampicin resistance was obtained in 3 (3.33%) cases (2 pulmonary, 1 extrapulmonary). Three indeterminate samples were found to be rifampicin sensitive on retesting.

Table 1. Demographic Data and Study Summary

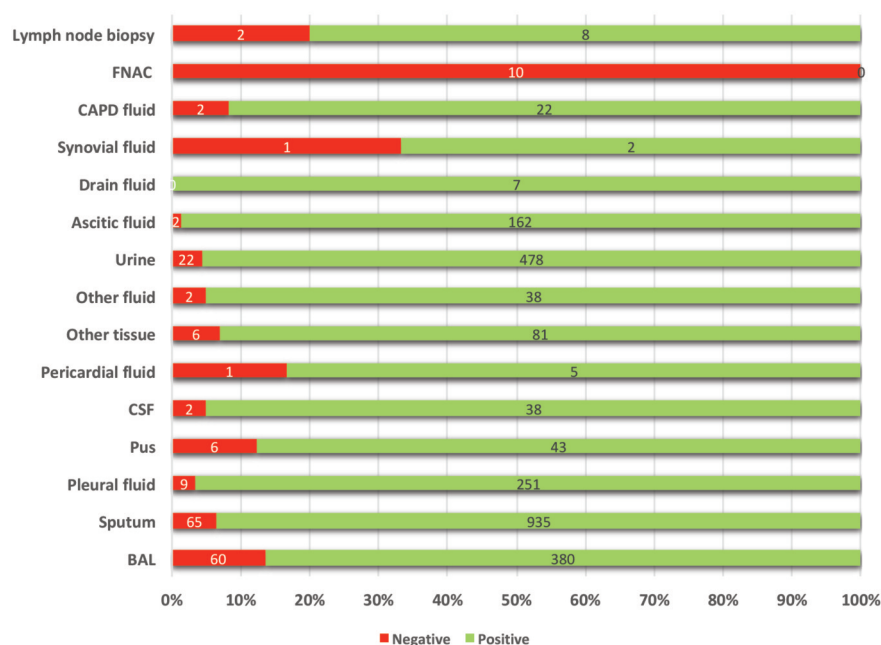
Variable	Category	No. of Patients (n = 2640)	Percentage of Total
Sex	Male	1760	66.67%
	Female	880	33.33%
Age categories, y	>15	10	0.39%
	15-29	625	23.67%
	30-44	979	37.08%
	45-59	525	19.89%
	60-74	480	18.18%
	>74	21	0.79%
MTB result	Detected	190	7.19%
	Not detected	2450	92.81%
Types of infection	Pulmonary	125	4.73%
	Extrapulmonary	65	2.46%
RR-MTB test result	RR-MTB detected	12	0.45%
	RR-MTB not detected	178	6.74%

Abbreviations: MTB, *Mycobacterium tuberculosis*; RR, rifampicin resistant

Table 2 shows baseline details of the KTR patients. Of the 12 patients diagnosed with RR-TB, there were 8 male patients and 4 female patients (male-to-female

ratio, 2:1). The median (with IQR) age of the RR group was 37 years (IQR, 26.2- 42.5 years). Of these 12 patients, 6 (50%) had history of antirejection therapy. One patient was already on continuation phase of antituberculosis treatment, and 2 patients were on intensive phase at the time of detection of MDR-TB. Analysis of the RR mutation pattern revealed *rpoB* gene mutations in 11 cases (81 bp rifampicin resistance determining region of MTB) and mutation combination, ie, mutations associated with more than 1 probe (probe D and probe E both) in 1 case. The most common *rpoB* mutations were located in the region of probe E 8/12 (75.0%), followed by probe A 2/12 (16.66%). The *rpoB* mutations were not seen in probe B and probe C. A combination of 2 probes, ie, probe D and probe E ("probe DE"), was seen in 1 case (8.33%). Table 3 shows the management protocol, outcomes, and follow-up of the cohort. In terms of treatment protocol, half of the patients required a bedaquiline-based short course treatment and half required a long course of the MDR-TB treatment regimen. The duration of treatment varied from 10 to 22 months. There were 3 biopsy-proven acute rejection episodes, which were managed successfully. One graft loss was reported in a patient who had baseline serum creatinine of 4 mg / dL. Two cases were lost to follow-

Figure 1. *Mycobacterium tuberculosis* Detected in Different Samples in Study Group



Abbreviations: BAL, bronchoalveolar lavage; CAPD, continuous ambulatory peritoneal dialysis; CSF, cerebrospinal fluid; FNAC, fine needle aspiration cytology

Table 2. Demographics, Baseline Details, Tuberculosis Type, and Microbiological Details of Resistance

Characteristic	Patient No.											
	1	2	3	4	5	6	7	8	9	10	11	12
Age, y	50	42	35	55	21	20	30	39	39	19	28	44
Sex	F	M	M	M	M	F	M	M	M	F	M	F
Blood group	A	B	B	A	B	O	O	AB	O	O	B	B
Native kidney disease	DM	IgAN	CGN	DM	CGN	VUR	FSGS	CIN	DM	CGN	UE	HTN
HIV/HBV/HCV reactive	N	N	N	N	HCV	N	N	HCV	N	N	N	HBV
Transplant type	LD	LD	LD	DD	LD	LD	LD	LD	DD	LD	LD	LD
Transplant age, y	3	2	7	1	0.6	1	2	3	10	1	2	3
Induction	ATG	ATG	IL-2RB	None	IL-2RB	ATG	IL-2RB	None	ATG	IL-2RB	IL-2RB	ATG
Tuberculosis type	PTB	PTB	PTB	PTB	PTB	PTB	PTB	PTB	PTB	PTB	GUTB	PTB
Sample	BAL	Sputum	BAL	BAL	BAL	Sputum	BAL	Sputum	BAL	Sputum	BAL	BAL
rpoB mutation												
Probe A	N	N	P	P	P	P	P	P	P	P	P	P
Probe B	P	P	P	P	P	P	P	P	P	P	P	P
Probe C	P	P	P	P	P	P	P	P	P	P	P	P
Probe D	P	P	P	N	P	P	P	P	P	P	P	P
Probe E	P	P	N	N	N	N	N	N	N	N	N	N

Abbreviations: ATG, antithymocyte globulin; BAL, bronchoalveolar lavage; CGN, chronic glomerulonephritis; CIN, chronic interstitial nephritis; DD, deceased donor; DM, diabetes; F, female; FSGS, focal segmental glomerulosclerosis; GUTB, genitourinary tuberculosis; HBV, hepatitis B virus; HCV, hepatitis C virus; HIV, human immunodeficiency virus; HTN, hypertension; IgAN, immunoglobulin A nephropathy; IL-2RB, interleukin-2 receptor blocker; LD, living donor; M, male; N, negative; P, positive; PTB, pulmonary tuberculosis; UE, unknown etiology; VUR, vesicoureteral reflex

Table 3. Baseline Drugs, Initial Treatment Protocol of Multidrug-Resistant Tuberculosis Regimen, Follow-Up Treatment Protocol, Outcome in Terms Of Survival and Graft Function, Follow-Up Duration

Patient	Baseline IS	Baseline ATT	Initial MDR-TB Regimen (duration, mo)	Conversion	Follow-up, y	Graft	Outcome
1	S+TAC+MMF	HRZE1	(18) Lfx, Bdq; (6) Lzd, # Cfz, Cs	NA	2	AR	LTF
2	S+TAC+MMF	HRZE2	(19) Lfx, Bdq; (6) Lzd, # Cfz, Cs	NA	3	AR	Cured
3	S+TAC	NA	(6) Bdq; (6) Lfx, Cfz, Z, E, Hh, Eto; (5) Lfx, Cfz, Z, E	NA	3	AR	Cured
4	S+EVE+MMF	NA	(4) Bdq; (6) Lfx, Cfz, Z, E, Hh, Eto; (5) Lfx, Cfz, Z, E	(18) Lfx, Bdq; (6) Lzd, # Cfz, Cs	1	Normal	Died
5	S+TAC+AZA	NA	(4) Bdq; (6) Lfx, Cfz, Z, E, Hh, Eto; (5) Lfx, Cfz, Z, E	(18) Lfx, Bdq; (6) Lzd, # Cfz, Cs	1	Normal	Died
6	S+TAC+MMF	NA	(6) Bdq; (6) Lfx, Cfz, Z, E, Hh, Eto; (5) Lfx, Cfz, Z, E	NA	3	Normal	Cured
7	S+TAC+MMF	NA	(4) Bdq; (6) Lfx, Cfz, Z, E, Hh, Eto; (5) Lfx, Cfz, Z, E	NA	4.1	Normal	Cured
8	S+CYC+MMF	NA	(4) Bdq; (6) Lfx, Cfz, Z, E, Hh, Eto; (5) Lfx, Cfz, Z, E	(19) Lfx, Bdq; (6) Lzd, # Cfz, Cs	0.6	Graft loss	LTF
9	S+TAC	HRZ1	(19) Lfx, Bdq; (6) Lzd, # Cfz, Cs	NA	2	Normal	Cured
10	S+TAC+AZA	NA	(4) Bdq; (6) Lfx, Cfz, Z, E, Hh, Eto; (5) Lfx, Cfz, Z, E	NA	1.5	Normal	Cured

Table 3. (Continued). Baseline Drugs, Initial Treatment Protocol of Multidrug-Resistant Tuberculosis Regimen, Follow-Up Treatment Protocol, Outcome in Terms Of Survival and Graft Function, Follow-Up Duration

Patient	Baseline IS	Baseline ATT	Initial MDR-TB Regimen (duration, mo)	Conversion	Follow-up, y	Graft	Outcome
11	S+TAC+MMF	NA	(6) Bdq; (6), Lfx, Cfz, Z, E, Hh, Eto; (5) Lfx, Cfz, Z, E	NA	0.3	Normal	Died
12	S+TAC+MMF	NA	(6) Bdq; (6) Lfx, Cfz, Z, E, Hh, Eto; (5) Lfx, Cfz, Z, E	NA	1.5	AR	Cured

Abbreviations: AR, acute rejection; ATT, antituberculosis treatment; AZA, azathioprine; Bdq, bedaquiline; Cfz, clofazimine; CYC, cyclosporine; E, ethambutol; Eto, ethionamide; EVE, everolimus; H, isoniazid; Hh, high-dose isoniazid; HRZE1 and HRZE2, isoniazid + rifampin + pyrazinamide + ethambutol for 1 mo or 2 mo, respectively; Lfx, levofloxacin; LTF, loss to follow-up; Lzd, linezolid; IS, immunosuppression; MDR-TB, multidrug-resistant tuberculosis; MMF, mycophenolic acid; NA, not available; Pdx, pyridoxine; R, rifampicin; S, steroid; TAC, tacrolimus; Z, pyrazinamide.

#Dose of linezolid was reduced. The numbers in parenthesis accompanying drug abbreviations are reported in months. Follow-up is reported in years.

up, and 3 patients died during the study. The complications associated with treatment are described in Table 4. Overall, the complications were infrequent and manageable.

Table 4. Complications During Treatment Protocol

Adverse Event	No. of Patients (% of total)
QT prolongation	0 (0%)
Seizures	3 (0.25%)
Skin rash	3 (0.25%)
Gastrointestinal symptoms	6 (0.5%)
Hepatitis	6 (0.5%)
Tendon rupture	1 (0.08%)
Psychotic symptoms	3 (0.25%)
Suicidal ideation	2 (0.17%)
Alopecia	3 (0.25%)
Gynecomastia	1 (0.08%)
Metallic taste	4 (0.33%)
Optic neuritis	1 (0.08%)
Tinnitus	5 (0.42%)
Giddiness	4 (0.33%)
Hypothyroidism	2 (0.17%)
Arthralgia	4 (0.33%)
Peripheral neuropathy	3 (0.25%)
Headache	6 (0.5%)
Depression	8 (0.67%)
Deranged sugars	3 (0.25%)
Superficial fungal infection	2 (0.17%)

Discussion

India ranks first among 22 countries with the highest incidence of TB, but the reliable information on the magnitude of MDR-TB in the country is still largely unavailable.¹² Studies detailing patterns of TB resistance in solid-organ transplant recipients have not been documented. In this study, we looked for the prevalence of TB and RR-TB in KTR patients along with frequency of specific mutations

responsible for RR by the GeneXpert assay. We opine that the GeneXpert assay can be used to diagnose RR-TB in KTR patients. The proportion of successful results with the GeneXpert assay was 97.7%, which is higher than study by Agizew and colleagues, who reported a successful result of 92.8%.¹³ In previously published studies, the most common error was the probe check control failure error. Improved technician skill for sample processing can overcome this error.¹⁴ Our study also shows that the GeneXpert assay provided successful results in extrapulmonary samples (such as stool, pleural / ascitic / pericardial / cerebrospinal fluids, pus, bone marrow, fine needle aspiration cytology, urine, and tissue biopsy), although the GeneXpert assay was designed for pulmonary samples.

We detected MTB in 7.9% of patients in our study, which is lower than the estimated data from the World Health Organization on RR in MTB.⁷ The proportion of RR-TB detection in our study was 12 (6.3%). Due to the paucity of studies in KTR patients, we compared our data with RR-TB in the general population (Table 5). Compared with the study from Jaipur conducted by Verma and colleagues, the prevalence of RR-TB in our study was lower (7.9% vs 11.09%, respectively).¹⁵ A higher prevalence of RR-TB has been reported in other studies from India (Lucknow 27.8%,¹⁶ New Delhi 17.9%,¹⁷ Central India 17%¹⁸). Possible causative factors could be differences in geography/location, methods of diagnosis, and TB control practices. We reported 12 samples with RR (11 cases with *rpoB* gene mutations and 1 case with mutation combination, both for

Table 5. Summary of Previous Studies (General Population)

Study (combination)	Country	Total	Probe				
			A	B	C	D	E
Ullah et al, 2016 (5)	Pakistan	408	5	44	6	34	314
Rahman et al, 2016 (0)	Bangladesh	91	2	14	3	13	59
Kaur et al, 2016 (1)	India 1	130	10	27	1	18	73
Reddy et al, 2017 (5)	India 2	171	14	26	1	31	94
Ochang et al, 2016 (3)	Nigeria	58	2	8	0	10	35
Mboowa et al, 2014 (0)	Uganda	12	1	3	0	1	7
Metcalfe et al, 2016 (0)	Zimbabwe	43	2	6	5	4	26
Present study, 2022 (1)	India	12	2	0	0	1	9

probe D and E, 8.33%). Prevalence of the combination mutation was less in our study compared with the study by Singhal and colleagues (6 strains [6/366] with more than 1 mutation).¹⁷ Mboowa and colleagues reported no sample with more than 1 probe failure (mutation combination).¹⁹ The location of the most common rpoB mutations reported by us was in the region of probe E with 8/12 (75.0%), followed by probe A with 2/12 (16.66%), and with absence of rpoB mutations in combination probe BC.

Indian studies have historically shown that the most common mutation conferring RR is located in the codon 531 of the rpoB gene. This observation is similar in KTR patients, as observed in our study. These studies also reported locations of other common rpoB mutations in the codon 516 in the probe A region and in the codon 526 of the probe D region. The implementation of the management protocol is difficult, as the treatment regimen has several adverse side effects, as well as drug-drug interaction. This required regular monitoring of blood counts, platelet counts, liver function tests, blood sugars, serum electrolytes, serum magnesium, serum calcium, Thyroid stimulating hormone, vision test, QT length, psychiatric evaluation, and audiometry tests. All patients on pantoprazole and domperidone combination were shifted to ranitidine. Similarly, the choice of antibiotics and antifungals was adjusted in case of any illness. Those infections comprised acute gastroenteritis and fungal infection in 2 patients each.

Overall, the outcome was acceptable, with mortality reported for 3 cases only. Similarly, the graft functioning was also acceptable for the cohort, with acute rejection reported in 4 cases and graft loss reported in only 1 case.

Limitations

We did not perform mycobacterial culture or anti-tubercular drug sensitivity tests, which are

considered the gold standard techniques, limiting in estimation of the proportion of false-positive and false-negative results of the GeneXpert assay. We also did not perform the rpoB gene sequencing process to establish the specific rpoB mutations that could have facilitated determination of the specificity and sensibility of the assay to detect mutations in the rpoB gene.²⁰ The treatment and follow-up details are retrospectively reviewed from memory as recalled by concerned doctors and from case files of patients.

Conclusions

The GeneXpert assay is a better screening test for rapid, early detection of MTB and RR and its further transmission in KTR patients. Additionally, it also is a useful technique to determine the mutation pattern of RR in MTB, which can help to understand the disease epidemiology and identification of the hot spots for implementation of TB control programs.

References

1. Yousef AI, Ismael MF, Elshora AE, Abdou HE. Pulmonary tuberculosis in patients with chronic renal failure at Zagazig University Hospitals. *Egypt J Chest Dis Tuberc*. 2014;63(1):187-192.
2. Zhang XF, Lv Y, Xue WJ, et al. Mycobacterium tuberculosis infection in solid organ transplant recipients: experience from a single center in China. *Transplant Proc*. 2008;40(5):1382-1385. doi:10.1016/j.transproceed.2008.01.075
3. Bosch W, Poowanawittayakom N, Chaikriangkrai K, et al. Tuberculous hepatitis in renal transplant recipients following alemtuzumab induction therapy. *Transpl Infect Dis*. 2013;15(1):E33-E39. doi:10.1111/tid.12048
4. Currie AC, Knight SR, Morris PJ. Tuberculosis in renal transplant recipients: the evidence for prophylaxis. *Transplantation*. 2010;90(7):695-704. doi:10.1097/TP.0b013e3181ecea8d
5. Grosset J. The sterilizing value of rifampicin and pyrazinamide in experimental short-course chemotherapy. *Bull Int Union Tuberc*. 1978;53(1):5-12.
6. World Health Organization. Companion Handbook to the WHO Guidelines for the Programmatic Management of Drug-Resistant Tuberculosis. Geneva: World Health Organization; 2014. <https://www.ncbi.nlm.nih.gov/books/NBK247420/>
7. World Health Organization. Global Tuberculosis Report 2016. Geneva: World Health Organization; 2016. <https://www.who.int/teams/global-tuberculosis-programme/tb-reports>

8. Lawn SD, Nicol MP. Xpert MTB/RIF assay: development, evaluation and implementation of a new rapid molecular diagnostic for tuberculosis and rifampicin resistance. *Future Microbiol.* 2011;6(9):1067-1082. doi:10.2217/fmb.11.84
9. World Health Organization. Automated real-time nucleic acid amplification technology for rapid and simultaneous detection of tuberculosis and rifampicin resistance: Xpert MTB/RIF assay for the diagnosis of pulmonary and extrapulmonary TB in adults and children: policy update. Geneva: World Health Organization; 2013. <https://apps.who.int/iris/handle/10665/112472>
10. Bodmer T, Ströhle A. Diagnosing pulmonary tuberculosis with the Xpert MTB/RIF test. *J Vis Exp.* 2012;(62):e3547. doi:10.3791/3547
11. Raizada N, Sachdeva KS, Sreenivas A, et al. Feasibility of decentralised deployment of Xpert MTB/RIF test at lower level of health system in India. *PLoS ONE.* 2014;9(2):e89301. doi:10.1371/journal.pone.0089301
12. World Health Organization. Global tuberculosis control: epidemiology, planning, financing: WHO report 2009. Geneva: World Health Organization; 2009. <https://apps.who.int/iris/handle/10665/44035>
13. Agizew T, Boyd R, Ndwapu N, et al. Peripheral clinic versus centralized laboratory-based Xpert MTB/RIF performance: experience gained from a pragmatic, stepped-wedge trial in Botswana. *PLoS One.* 2017;12(8):e0183237. doi:10.1371/journal.pone.0183237.
14. Creswell J, Codlin AJ, Andre E, et al. Results from early programmatic implementation of Xpert MTB/RIF testing in nine countries. *BMC Infect Dis.* 2014;14(2). doi:10.1186/1471-2334-14-2
15. Verma D, Malhotra B, Goyal S, Shah T, Gupta KN. Prevalence of MDR-TB and mutational pattern in *rpoB* gene of *Mycobacterium tuberculosis* at Eastern Rajasthan. Microcon 2014. 38th National Conference of Indian Association of Medical Microbiologists. 2014;MP39:253.
16. Jain A, Diwakar P, Singh U. Declining trend of resistance to first-line anti-tubercular drugs in clinical isolates of *Mycobacterium tuberculosis* in a tertiary care north Indian hospital after implementation of revised national tuberculosis control programme. *Indian J Med Microbiol.* 2014;32(4):430-433. doi:10.4103/0255-0857.142257.
17. Singhal R, Myneedu VP, Arora J, et al. Early detection of multi-drug resistance and common mutations in *Mycobacterium tuberculosis* isolates from Delhi using GenoType MTBDRplus assay. *Indian J Med Microbiol.* 2015;33 Suppl:46-52. doi:10.4103/0255-0857.150879
18. Desikan P, Chauhan DS, Sharma P, Panwalkar N, Yadav P, Ohri BS. Clonal diversity and drug resistance in *Mycobacterium tuberculosis* isolated from extra-pulmonary samples in central India: a pilot study. *Indian J Med Microbiol.* 2014;32(4):434-437. doi:10.4103/0255-0857.142255
19. Mboowa G, Namaganda C, Ssengooba W. Rifampicin resistance mutations in the 81 bp RRDR of *rpoB* gene in *Mycobacterium tuberculosis* clinical isolates using Xpert MTB/RIF in Kampala, Uganda: a retrospective study. *BMC Infect Dis.* 2014;14:481. doi:10.1186/1471-2334-14-481
20. Suresh N, Singh UB, Arora J, et al. *rpoB* gene sequencing and spoligotyping of multidrug-resistant *Mycobacterium tuberculosis* isolates from India. *Infect Genet Evol.* 2006;6(6):474-483. doi:10.1016/j.meegid.2006.03.001